

SM015

Mathematics 1
Pre PSPM
Semester I
Session 2024/25

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Matematik 1
Pra PSPM
Semester I
Sesi 2024/25



UNIT MATEMATIK
KOLEJ MATRIKULASI LABUAN

PRA PSPM SET 1

NAMA	
KELAS	
NO. MATRIK	
NAMA PENSYARAH	

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** the question given.

1 Evaluate,

a) $\lim_{x \rightarrow -2} \left(\frac{x^3 + 8}{2x^2 + 4x} \right)$

[3 marks]

b) $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{3x^2 + 7x}}{2x + 5} \right)$

[4 marks]

c) $\lim_{x \rightarrow 3^-} \left(\frac{x^2 + 2x - 15}{|x - 3|} \right)$

[3 marks]

2 Given $y = \sqrt{1 - 2x}$,

a) By using the first principle, find $\frac{dy}{dx}$.

[5 marks]

b) Hence, find $\frac{d^2y}{dx^2}$.

[2 marks]

3 By considering the first and derivatives of $f(x) = \frac{\ln(x-2)}{2-x}$ for $x > 2$, show that there is only one minimum point.

[5 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** the question given.

1. Given $z = \frac{\sqrt{2} - i}{1 - i\sqrt{2}}$. Express z in the form of $a + bi$. Hence, determine the modulus and argument of its conjugate, $\bar{z}$.

[6 marks]

2. a) Solve ,

$$\log_2 p + \log_8 2p = -1$$

[4 marks]

- b) Find the solution set of $\left| \frac{x-1}{3x+2} \right| \leq 2$.

[7 marks]

3. Given $f(x) = \left(\frac{3x}{5x+1} \right)$ and $g(x) = \frac{x}{\sqrt{x^2-4}}$. Find,

- a) The domain of $g(x)$.

[3 marks]

- b) $h(x)$, if $(f \circ h)(x) = x$.

[4 marks]

4. Given a function $f(x) = \ln(3x+6)$,

a) Show that $f(x)$ is a one to one function by using algebraic approach.

[2 marks]

b) Find the inverse function of $f(x)$ and state its domain and range.

Hence find the value of x for which $f^{-1}(x) = 0$.

[7 marks]

c) Sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same coordinate axes.

[3 marks]

5 a) The function f is defined by,

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & , x < 1 \\ m^2-2 & , x = 1 \\ \frac{|1-x|}{x-1} + n & , x > 1 \end{cases}$$

Where m and n are constants.

i. If $\lim_{x \rightarrow 1} f(x)$ is exist, find the value of n .

[5 marks]

ii. If f is continuous at $x=1$, determine the values of m .

[2 marks]

b) A function f is defined by $f(x) = \frac{x}{x+3}$. Find all the asymptotes of f . Hence, sketch the graph and state the domain of f .

[7 marks]

6 a) If $xy = 2(x - y)^2$, find the following values at $\left(1, \frac{1}{2}\right)$.

i) $\frac{dy}{dx}$

ii) $\frac{d^2y}{dx^2}$

(9 marks)

b) Given $x = \cos 2t$ and $y = 2t + \sin^2 t$, show that $\frac{dy}{dx} = -\operatorname{cosec} 2t - \frac{1}{2}$ and evaluate $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{8}$.

(7 marks)

7 A cuboid has a rectangular base length of $2x$ cm, width x cm and height y cm. If the volume of the cuboid is 72cm^3 .

a) Show the surface area $A \text{ cm}^2$, is given by,

$$A = 4x^2 + \frac{216}{x}$$

[3 marks]

b) Determine the value x for which the surface area of the cuboid is minimum. Hence, find the minimum surface area of the cuboid.

[6 marks]

END OF QUESTION PAPER